Unsupervised Deformable Bilinear Fusion Network for Unregistered Hyperspectral Image Super-Resolution

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***Abstract*—Hyperspectral image super-resolution (HSI-SR) aims to reconstruct a high-resolution hyperspectral image (HrHSI) by fusing a low-resolution hyperspectral image (HSI) with a high-resolution multispectral image (MSI). In practical HSI–MSI fusion, spatial misregistration and nonlinear spectral mixing often occur simultaneously, whereas many existing methods assume accurate registration or rely mainly on linear mixing models (LMMs). To address these issues, we propose an unsupervised deformable bilinear fusion network (UDBFN). The UDBFN integrates a hierarchical registration module (HRM) for multiscale abundance alignment, a cross-modality Transformer (CMT) for estimating cross-modal displacement fields, and a bilinear decoder for modeling linear and pairwise nonlinear mixing components. Experiments on three benchmark datasets show that the UDBFN provides competitive reconstruction accuracy under well-registered settings and is particularly robust under rotation, translation, and nonrigid misregistration. The code is available at https://github.com/zoukeh/UMDBN-master**

***Index Terms*—Bilinear unmixing, hyperspectral image (HSI)–multispectral image (MSI) fusion, hyperspectral image super-resolution (HSI-SR), image registration, Transformer.**

NOMENCLATURE

X	Low-resolution HSI (LrHSI).
Y	High-resolution MSI (HrMSI).
Z	Target high-resolution HSI (HrHSI).
P	Point spread function (PSF).
Q	Spectral response function (SRF).
A	Endmember matrix.
S	Abundance matrix.
B	Bilinear coefficient matrix.
A_X	Endmember matrix estimated from the HSI.
A_Y	Endmember matrix estimated from the MSI.
S_X	Abundance extracted from the LrHSI encoder.
S_Y	Abundance extracted from the HrMSI encoder.
B_X	Bilinear coefficient matrix from the HSI.
B_Y	Bilinear coefficient matrix from MSI.
$\hat{S}_X$	Registered abundance map from the HSI after the HRM.
$H(A)$	Hadamard product of endmembers.
$\hat{H}(A)$	Transformed Hadamard product for the MSI.
φ_i	Displacement field at scale i in the HRM.
N	Number of scales in the HRM.
α, β, γ	Tradeoff parameters in the loss function.
L_{rec}	Reconstruction loss.
L_{cons}	Abundance sum–to–one constraint loss.
L_{spa}	Abundance sparsity loss.
L_{reg}	Registration loss.

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I. INTRODUCTION

HYPERSPECTRAL images (HSIs) contain rich spectral information but often have limited spatial resolution because of sensor and imaging constraints [1]. Multispectral images (MSIs), by contrast, provide higher spatial resolution with fewer spectral bands. HSI super-resolution (HSI-SR) therefore commonly fuses a low-resolution HSI (LrHSI) and a high-resolution MSI (HrMSI) of the same scene to recover a high-resolution HSI (HrHSI), which can further benefit downstream remote-sensing tasks [2].

According to the recent survey by Wang et al. [3], MSI-guided HSI super-resolution (HSI-SR) methods can be broadly grouped into model-based methods, decomposition-based methods, and deep network-based methods. Model-based approaches use degradation models and handcrafted priors, decomposition-based approaches exploit spectral unmixing theory, and deep network-based approaches learn spatial–spectral mappings with architectures such as CNNs and Transformers. These categories provide useful accuracy and interpretability tradeoffs, but they also reveal two practical difficulties: multimodal observations are not always perfectly registered, and real scenes may violate a purely linear mixing assumption.

Model-based methods improve the spatial resolution of the low-resolution HSI (LrHSI) by constructing an objective function from spatial and spectral priors and then solving the resulting optimization problem [4]. Wei et al. [5] used a half-quadratic splitting strategy to decompose the optimization into subproblems. Dian et al. [6] incorporated a CNN denoiser into an ADMM framework to estimate sparse coefficients. Wu et al. [7] proposed CSRNet to constrain HSI and MSI restoration branches mutually. Related deep-prior methods [8],

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[9] further combine physical degradation models with data-driven regularization. These approaches are interpretable, but their performance depends strongly on the accuracy of the assumed degradation model.

Decomposition-based methods assume that an HSI can be represented by endmembers and abundances [10]. Because HrHSI and HrMSI share abundance information, while HrHSI and LrHSI share spectral endmembers, these methods can exploit physically meaningful shared variables. Dictionary and matrix-factorization methods have also been widely studied. For example, optimized twin dictionaries improve coupled sparse representation for HSI–MSI fusion [11], and dictionary expansion has recently been used to handle incompletely overlapped multi- and hyperspectral observations [12]. These methods provide strong interpretability, but they often rely on accurate dictionary learning, degradation calibration, or overlap assumptions.

Deep-learning-based methods learn the mapping from LrHSI and HrMSI to HrHSI directly. Early CNN-based fusion methods [13], [14] were followed by multiscale networks that exploit information at different spatial resolutions [15]. Recent Transformer-based methods further model long-range spatial–spectral dependencies and cross-modal interactions [16], [17]. These methods are flexible and effective. Their robustness can still degrade when the two modalities are spatially misaligned or when nonlinear spectral interactions remain outside the model.

Recent unregistered HSI-SR approaches attempt to address spatial inconsistency by incorporating deformable alignment or mutual information constraints [18], [19], [20]. However, many of them still rely on simplified linear mixing assumptions and therefore have limited ability to represent nonlinear interactions such as multiple scattering among endmembers.

In this work, we focus on fusion-based HSI-SR under misregistration, where an HrMSI assists the recovery of an HrHSI. To jointly address cross-modal spatial deformation and nonlinear spectral mixing, we propose an unsupervised deformable bilinear fusion network (UDBFN). Our main contributions are as follows.

1) We propose UDBFN, an unsupervised HSI–MSI fusion framework that couples cross-modal deformable registration with bilinear spectral mixing. The method targets misregistered fusion scenarios, with an HrMSI assisting HSI reconstruction.
2) We design a hierarchical registration module (HRM) with a cross-modality Transformer (CMT) to estimate multiscale displacement fields in the abundance domain, and we use a bilinear decoder to represent both linear components and pairwise nonlinear interactions. Experiments indicate that the main advantage of UDBFN lies in its robustness to rotation, translation, and nonrigid misregistration while maintaining competitive accuracy under well-registered settings.

Here, "deformable" in the proposed UDBFN refers to the HRM-based registration component that handles cross-modal spatial deformation; the complete contribution of UDBFN includes this deformable alignment together with bilinear spectral mixing and unsupervised HSI–MSI fusion.

TABLE I
COMPARISON OF RELATED FUSION CONCEPTS

Method type	Unreg.	Hier. align.	Bilinear	Unsuper.
CNMF/CO-CNMF [29], [26]	–	–	–	Yes
R-FUSE/HySure [27], [30]	–	–	–	Yes
Dictionary fusion [12], [11]	Partial	–	–	Yes
NonRegSRNet/XINet [19], [22]	Yes	Partial	–	Yes
Transformer fusion [16], [25]	Partial	Yes	–	Yes
UDBFN	Yes	Yes	Yes	Yes

II. Related Work

A. HSI-SR With Registration

Due to the differences between multimodal sensors, there is often some deviation between HSIs and MSIs, which can cause degradation of super-resolution effects [18]. u^{2}MDN [21] encourages extracted spatial representations to satisfy Dirichlet-distribution constraints and can handle rigid and nonrigid deformations. Approaches such as XINet [22], STN-based methods [23], and NonRegSRNet [19] exploit feature alignment, cross-modality learning, affine transformation, or nonrigid registration to improve fusion robustness. Lai et al. [20] estimated dense optical flow between hyperspectral and multispectral features for real unaligned guidance.

More recently, DTDNML [24] formulates unsupervised blind HSI–MSI fusion via deep Tucker decomposition and spatial–spectral manifold learning, and UMSFT [25] introduces a multilevel spatial–spectral Transformer. Classical and unrolled fusion models such as CO-CNMF [26], R-FUSE [27], and NMF-DuNet [28] also improve coupled matrix factorization or optimization-based fusion. However, these methods are mainly designed for standard fusion settings and generally assume either reliable degradation models or well-aligned multimodal observations. Table I summarizes the relationship between representative concepts and our design. UDBFN's novelty comes from coupling hierarchical cross-modal abundance registration with a bilinear decoder in an unsupervised HSI–MSI fusion framework; the term "deformable" only describes one component of this design.

In conclusion, model-driven approaches rely on handcrafted priors such as sparsity or low-rankness, which oversimplify real-world spectral complexity and hinder the recovery of fine spatial details. Single-image SR models, though end-to-end trainable, assume ideal degradations and generalize poorly under complex conditions. Fusion-based frameworks often suffer from spectral distortion caused by misregistration and typically separate registration from fusion, leading to cumulative errors. Even recent unsupervised or decomposition-based strategies depend on linear mixing assumptions and accurate degradation models, limiting real-world applicability. These challenges motivate a unified, registration-aware framework capable of handling unregistered multimodal inputs while preserving spectral fidelity.

B. Spectral Mixing Model

In spectral mixing tasks, it is typically modeled by a linear mixing model (LMM), which represents each pixel as

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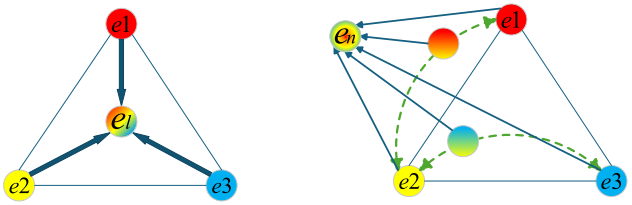


Fig. 1. Illustration of (a) linear and (b) nonlinear mixing. In linear mixing, the mixed pixel lies inside the simplex formed by endmembers, with straight arrows denoting direct linear contributions. In nonlinear mixing, the mixed pixel appears outside the simplex, and curved arrows indicate complex interactions such as multiple scattering.

a linear combination of end-member spectra weighted by its abundance. The LMM [31] assumes that the ground objects on the reflective surface are distributed in a chessboard pattern, and the incident light is directly received by the sensor after interacting with only one substance each time. It can be expressed as

$$z_j = \sum_{i=1}^{r} a_i s_{i,j} + \varepsilon_j \tag{1}$$

where $z_j \in \mathbb{R}^{C\times 1}$ represents the vector of the jth HSI pixel, a_i represents endmember, and $s_{i,j}$ represents coefficient of the endmember vector. ε_j represents the additional uncorrelated Gaussian noise.

Although LMM performs well in the instantaneous field of view (IFOV), its performance will be significantly reduced in complex natural scenarios due to many nonlinear scattering cases. Fig. 1(a) and (b) represents the case of a linear and nonlinear mixed scene, respectively. (The simplex interpretation under the LMM is an intuitive geometric illustration and generally relies on additional abundance constraints along with linearity.) Experiments [32] have shown that, using the local linear Gaussian process latent model (LL-GPLVM) [33] that based on nonlinear spatial representation and PCA, respectively, to represent HSI data, the average reconstruction error (ARE) of LL-GPLVM is lower than that of PCA, which indicates that the HSI is more suitable to be represented using nonlinear models.

The nonlinear mixing model (NMM) is capable of accounting for inherent nonlinear mixing effects, particularly in scenarios involving complex multilayered or intimate mixtures. However, the optimization problems associated with the NMM are often challenging to solve. In recent years, the bilinear mixing model (BMM), owing to its mathematically tractable form, has significantly advanced the development of nonlinear unmixing methods. Among representative BMMs [34], [35], [36], [37], the generalized bilinear model (GBM) and the polynomial postlinear model (PPNM) have emerged as widely adopted methods for nonlinear unmixing, because of their strong generalization capabilities.

In conclusion, despite progress in unregistered HSI-SR, many existing methods rely on linear mixing assumptions and treat registration as an auxiliary or loosely coupled step, making them vulnerable to spectral distortions, inaccurate abundance estimation, and nonrigid misalignments. In contrast, our framework tightly integrates an HRM for multiscale deformable alignment, a CMT for robust spectral–spatial correspondence, and a bilinear unmixing decoder to capture both linear and nonlinear mixing effects. This end-to-end unsupervised design ensures geometric and semantic consistency, improves registration accuracy, and enhances spectral fidelity and spatial detail, distinguishing our approach from prior works that oversimplify mixing or decouple registration and fusion.

TABLE II
PAVIA UNIVERSITY, WASHINGTON DC, AND CHIKUSEI DATASET INTRODUCTION

Dataset name	ratio	Data
Pavia University	8	HrHSI:256×256×103 HrMSI:256×256×4 LrHSI:32 × 32 ×103
Washington DC	4	HrHSI:256×256×191 HrMSI:256×256×6 LrHSI:64×64×191
Chikusei	8	HrHSI:256×256×112 HrMSI:256×256×4 LrHSI:32×32×112

III. PROPOSED METHOD

A. Problem Formulation

Given an LrHSI block $\boldsymbol{X} \in \mathbb{R}^{hw\times C}$ and an HrMSI $\boldsymbol{Y} \in \mathbb{R}^{HW\times c}$ captured over the same area, where $H > h$, $W > w$, $C > c$, and $H/h = W/w = r$, $\boldsymbol{X}$ and $\boldsymbol{Y}$ can be considered as the results of certain degradations of $\boldsymbol{Z} \in \mathbb{R}^{HW\times C}$ in the spatial and spectral domains, respectively. This relationship can be formulated based on the LMM

$$\boldsymbol{X} = \boldsymbol{PZ} + \mathcal{N}_{\mathcal{X}} \tag{2}$$

$$\boldsymbol{Y} = \boldsymbol{ZQ} + \mathcal{N}_{\mathcal{Y}} \tag{3}$$

where $\boldsymbol{P} \in \mathbb{R}^{hw\times HW}$ denotes the PSF, which represents a spatial decline from HrHSI to LrHSI. $\boldsymbol{Q} \in \mathbb{R}^{C\times c}$ is the SRF representing the degradation process of $\boldsymbol{Z}$ in the spectral domain. The noise $\mathcal{N}_{\mathcal{X}}$ and $\mathcal{N}_{\mathcal{Y}}$ caused by sensors during the imaging process will be overlooked in the following parts. Our objective is to reconstruct the HrHSI $\boldsymbol{Z}$ utilizing both $\boldsymbol{X}$ and $\boldsymbol{Y}$. So, our fusion model can be represented as

$$\min_{\boldsymbol{Z}} \|\boldsymbol{X} - \boldsymbol{PZ}\|_F^2 + \|\boldsymbol{Y} - \boldsymbol{ZQ}\|_F^2. \tag{4}$$

Specifically, the LrHSI is simulated by applying a Gaussian PSF with a standard deviation of 0.5 to the HrHSI, followed by downsampling. The scale factor is 8 for Pavia University and Chikusei and 4 for Washington DC, as listed in Table II. The corresponding MSI (HrMSI) is generated by multiplying the HrHSI with a predefined SRF provided by the dataset. All datasets use the same simulation procedure to ensure consistency across experiments.

The LMM simply unmixes spectral images into abundance and endmembers, neglecting the light scattering between endmembers [34]. The unmixed endmember and abundance

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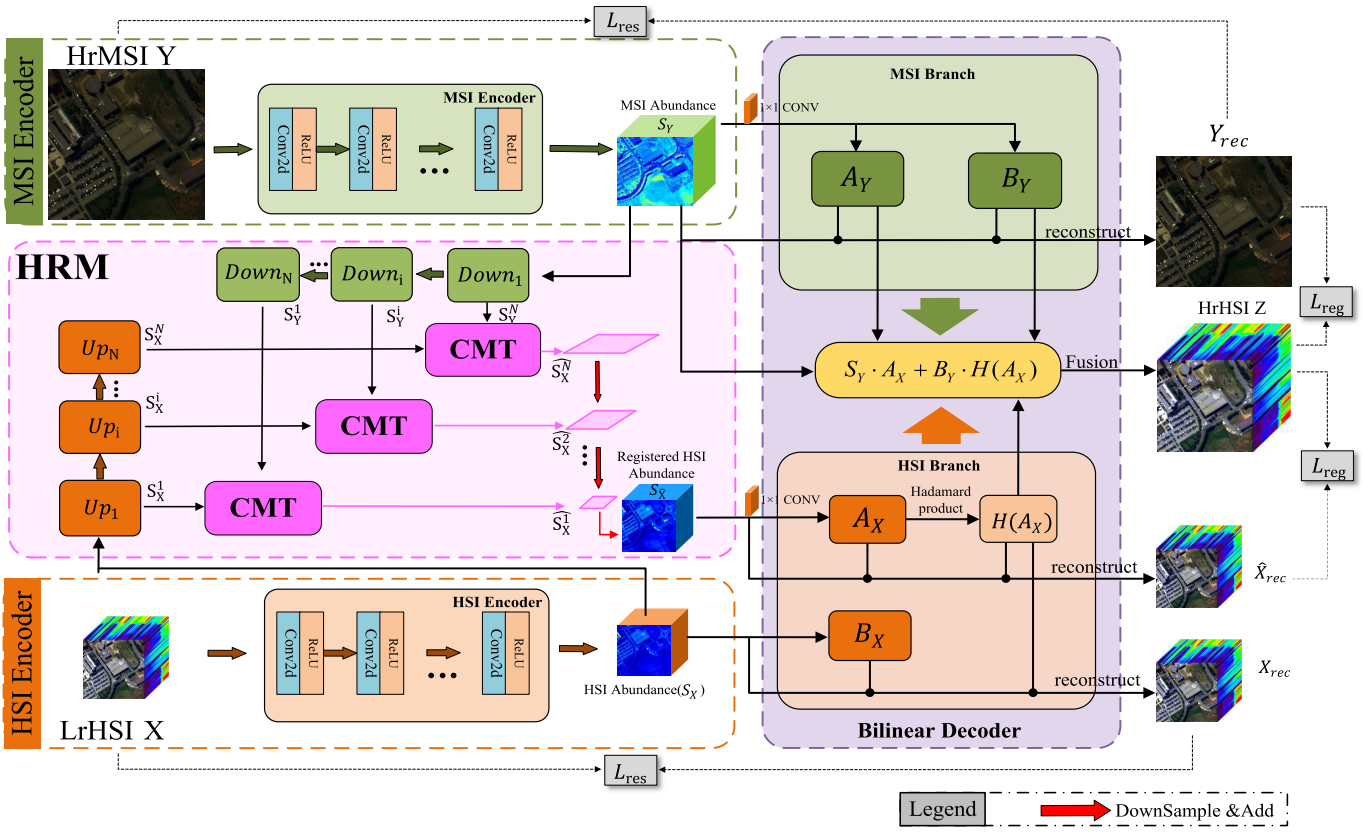


Fig. 2. Flowchart illustrating the general structure of our proposed HSI-SR method, which comprises three main modules: HSI and MSI encoders, HRM, and bilinear decoder. For the convenience of formula writing, we reverse the order of subscript naming in S_Y^i.

information may be distorted. The inaccurate unmixing of HSI's endmembers and MSI's abundance will reduce the restoration accuracy of the fused image. Therefore, it is necessary to consider the bilinear mixing effect caused by scattering between end elements. We can formulate the jth vector z_j in $\boldsymbol{Z}$ as

$$z_j = \sum_{i=1}^{n} a_i s_{i,j} + \sum_{i=1}^{n} \sum_{k=i}^{n} (a_i \odot a_k) b_{i,k,j}$$
$$s.t.\ s_{i,j}, b_{i,k,j} \geq 0, \quad 0 \leq j \leq HW \tag{5}$$

where given the abundance $\boldsymbol{S} = [s_1, s_2, \ldots, s_{HW}] \in \mathbb{R}^{HW \times n}$. $a_i, a_k \in \mathbb{R}^{C \times 1}$ represent endmember vectors in $\boldsymbol{A} = [a_1, a_2, \ldots, a_n] \in \mathbb{R}^{C \times n}$. We assume that there are n endmembers in $\boldsymbol{Z}$. $\odot$ is a Hadamard product operator that represents the mixed endmember information that interacts between every two endmember vectors (including itself). We stack all pairwise Hadamard products into a bilinear endmember matrix

$$H(\boldsymbol{A}) = [a_1 \odot a_1, \ldots, a_i \odot a_k, \ldots, a_n \odot a_n]^T \tag{6}$$

where $b_{i,k,j}$ are the elements in the parameter matrices $\boldsymbol{B} \in \mathbb{R}^{HW \times n(n+1)/2}$ which are the matrix of the coefficients of the bilinear scattering components. Equation (5) can be expressed in matrix notation as follows:

$$\boldsymbol{Z} = \boldsymbol{S} \cdot \boldsymbol{A} + \boldsymbol{B} \cdot H(\boldsymbol{A}) \tag{7}$$

where the dot operator denotes matrix multiplication.

Based on (2) and (3), our model can be represented as

$$\min_{\boldsymbol{Z}} \left\| \boldsymbol{X} - \boldsymbol{PS} \cdot \boldsymbol{A} - \boldsymbol{PB} \cdot H(\boldsymbol{A}) \right\|_2^2 + \left\| \boldsymbol{Y} - \boldsymbol{S} \cdot \boldsymbol{AQ} - \boldsymbol{B} \cdot H(\boldsymbol{A})\boldsymbol{Q} \right\|_2^2. \tag{8}$$

B. Overall Pipeline

To address the challenges mentioned, we propose a UDBFN for fusing unregistered LrHSI and HrMSI end-to-end to generate HrHSI. UDBFN consists of two spectral encoders (HSI and MSI encoders), a registration module, and a bilinear decoder, enabling accurate decomposition and reconstruction of hyperspectral and MSIs. Fig. 2 shows the framework of the UDBFN.

During training, the network uses only the observed LrHSI $\boldsymbol{X}$ and HrMSI $\boldsymbol{Y}$ as input pairs. The ground-truth HrHSI $\boldsymbol{Z}$ is not used as a supervisory label; optimization is driven by losses computed from $\boldsymbol{X}$, $\boldsymbol{Y}$, their reconstructions, and the cross-modal consistency terms defined in Section III-E. In the simulated experiments, $\boldsymbol{Z}$ is used only for generating degraded observations and for evaluation.

1) Spectral Encoder: In (8), we make the following variable substitutions to better distinguish:

$$\begin{cases} \boldsymbol{A_X} = \boldsymbol{A}, & \boldsymbol{S_X} = \boldsymbol{PS}, & \boldsymbol{B_X} = \boldsymbol{PB} \\ \boldsymbol{A_Y} = \boldsymbol{AQ}, & \boldsymbol{S_Y} = \boldsymbol{S}, & \boldsymbol{B_Y} = \boldsymbol{B}. \end{cases} \tag{9}$$

This shows that $\boldsymbol{X}$ retains the spectral information of $\boldsymbol{Z}$, while $\boldsymbol{Y}$ has partially lost the spectral information but still stores the

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original abundance information of $\boldsymbol{Z}$. We adopt HSI and MSI encoder branches f_{HSI} and f_{MSI} to extract abundance features, respectively. These two branches are composed of multiple 1×1 conv layers and a ReLU activation function stacked together. The process can be expressed as follows:

$$\boldsymbol{S_X} = f_{HSI}(\boldsymbol{X}), \quad \boldsymbol{S_Y} = f_{MSI}(\boldsymbol{Y}). \tag{10}$$

2) Hierarchical Registration Module: The registration module named HRM aligns the $\boldsymbol{S_X}$ and $\boldsymbol{S_Y}$ by capturing spatial correlations at multiple scales. The HRM generates the displacement fields to warp the moving feature $\boldsymbol{S_X}$. The output of the HRM is the registered hyperspectral abundance matrix $\widehat{\boldsymbol{S_X}}$. For detailed procedures, refer to Section III-C.

3) Bilinear Decoder: The bilinear decoder reconstructs the abundance $\boldsymbol{S_X}$, $\boldsymbol{S_Y}$, and $\widehat{\boldsymbol{S_X}}$ to the original images. $\boldsymbol{S_X}$ and $\boldsymbol{S_Y}$ are fed into HSI and MSI branches, respectively, denoted as d_{HSI} and d_{MSI}. $\widehat{\boldsymbol{S_X}}$ is also fed into the HSI branch to obtain the aligned LrHSI $\widehat{\boldsymbol{X_{rec}}}$. It can be formulated as

$$\begin{cases} \boldsymbol{X_{rec}} = d_{HSI}(\boldsymbol{S_X}; \boldsymbol{A_X}) \\ \widehat{\boldsymbol{X_{rec}}} = d_{HSI}\left(\widehat{\boldsymbol{S_X}}; \boldsymbol{A_X}\right) \\ \boldsymbol{Y_{rec}} = d_{MSI}(\boldsymbol{S_Y}; \boldsymbol{A_Y}) \end{cases} \tag{11}$$

where $\boldsymbol{A_X}$ and $\boldsymbol{A_Y}$ are parameters in d_{HSI} and d_{MSI}, respectively. The decoder learns the linear and bilinear mixed components of the LrHSI and HrMSI separately and shares the corresponding parameters between the two modalities to reconstruct $\boldsymbol{Z}$. Based on (7), it can be written as

$$\boldsymbol{Z} = \boldsymbol{S_Y}\boldsymbol{A_X} + \boldsymbol{B_Y}H(\boldsymbol{A_X}). \tag{12}$$

For detailed procedures, refer to Section III-D.

4) Loss Function: To optimize the parameters of the UDBFN, we define four loss functions, including L_{rec}, L_{cons}, L_{spa}, and L_{reg} in Section III-E. L_{rec} and L_{reg} are mainly used to minimize the gap between the reconstructed image and the input. L_{cons} and L_{spa} are constraints on the components.

C. Hierarchical Registration Module

In the registration task, HrMSI acts as the target image due to its detailed spatial data, while LrHSI is the moving image [38]. To address the issue of misalignment between LrHSI and HrMSI, we design a feature pyramid-based alignment module named the HRM, which is better able to adapt to changes in scenes of different scales. Given the significant difference in spatial resolution between LrHSI and HrMSI, direct alignment proves to be highly challenging. To address this issue, employing a hierarchical feature pyramid structure could be a solution. This approach enables the extraction of features across multiple scales, thereby facilitating more accurate alignment. The HRM uses a multilayer alignment structure to focus on spatial features in uniformly sized image pairs, consisting of N layers. This module captures features from both modalities at multiple scales, aligns them, and produces a calibrated abundance matrix. The process can be expressed as follows:

$$\begin{cases} \boldsymbol{S_Y^i} = Down_{N-i+1}\left(\boldsymbol{S_Y^{i-1}}\right) \\ \boldsymbol{S_X^i} = Up_i\left(\boldsymbol{S_X^{i-1}}\right) \end{cases} \tag{13}$$

where $\boldsymbol{S_Y^i}$ and $\boldsymbol{S_X^i}(i \in [1, N])$ represent the MSI and HSI abundance map at the ith scale, which is the same size, respectively. We use a convolution kernel with a stride of 2 for pooling and transposed convolution to upsample. Note that Up_1 and $Down_N$ do not perform reshape operations.

1) Cross-Modality Transformer:

$$\begin{cases} \boldsymbol{Q}^i = \boldsymbol{W}_q \boldsymbol{S_X^i} \\ \boldsymbol{K}^i = \boldsymbol{W}_k \boldsymbol{S_Y^i} \\ \boldsymbol{V}^i = \boldsymbol{W}_v \boldsymbol{S_Y^i} \end{cases} \tag{14}$$

where $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_v$ are the weight matrices of the dense layers. We compute the spatial attention matrix by

$$\boldsymbol{O}_i = \text{softmax}\left(\boldsymbol{Q}^i\left(\boldsymbol{K}^i\right)^T\right)\boldsymbol{V}^i. \tag{15}$$

We use $f_\theta(\cdot)$ to fit the deformation field. $f_\theta(\cdot)$ adopts a U-net structure with skip connections to estimate the displacement field $\boldsymbol{\phi}_i$, which can be formulated as

$$\boldsymbol{\phi}_i = f_\theta(\boldsymbol{O}_i) \tag{16}$$

where $\boldsymbol{\phi}_i \in \mathbb{R}^{h_i \times w_i \times 2}$ represents 2-D pixel offsets in $\boldsymbol{S_X^i}$. h_i and w_i are the height and width of the ith layer, respectively. To guide the deformation of the HSI feature map $\boldsymbol{S_X^i}$ based on $\boldsymbol{S_Y^i}$, we use a grid generator to produce a transformed grid and a sampler to interpolate the pixel values. We denote this spatial warping/sampling operation by $\mathcal{W}(\cdot, \cdot)$

$$\widehat{\boldsymbol{S}}_X^i = \mathcal{W}\left(\boldsymbol{S_X^i}, \boldsymbol{\phi}_i\right). \tag{17}$$

2) Registration Aggregation: Finally, we need to aggregate output information at different scales. As shown in Fig. 2, the process of aggregation can be formulated as

$$\widehat{\boldsymbol{S_X}} = f_{PAN}\left(\widehat{\boldsymbol{S_X^1}}, \widehat{\boldsymbol{S_X^2}}, \ldots, \widehat{\boldsymbol{S_X^N}}\right) \tag{18}$$

where f_{PAN} represents the process of hierarchical feature aggregation. Each $\widehat{\boldsymbol{S}}_X^i$ is the registered abundance map at the ith scale, obtained by warping $\boldsymbol{S_X^i}$ with the CMT-estimated displacement field. Larger registration features are concatenated with the next layer size through downsampling and reintegrated through a CNN. $\widehat{\boldsymbol{S_X}}$ is the final output of the HRM.

We use the CMT strategy for extracting multiscale features hierarchically, efficiently capturing HrMSI details and accommodating varying feature scales. The detailed architecture of CMT is illustrated in Fig. 3. Overall, the HRM addresses the spatial resolution disparity between LrHSI and HrMSI by aligning multiscale abundance maps from the two modalities.

D. Bilinear Decoder

Bilinear decoder has two decoder streams to reconstruct $\boldsymbol{S_X}$ and $\boldsymbol{S_Y}$ to the original input $\boldsymbol{X}$ and $\boldsymbol{Y}$, respectively. Each stream has linear and bilinear branches to calculate the linear and bilinear components in the mixed pixels. In the linear branch of HSIs, $\boldsymbol{S_X}$ is fed into a 1×1 convolutional layer without a bias variable, and the parameters of this layer are regarded as the endmembers matrix of HSI $\boldsymbol{A_X} \in \mathbb{R}^{C \times n}$. The MSI branch performs the same operation to obtain the endmembers matrix of MSI $\boldsymbol{A_Y} \in \mathbb{R}^{c \times n}$ from $\boldsymbol{S_Y}$. The bilinear branches construct bilinear components of HSIs and MSIs, enabling our method to represent pairwise nonlinear spectral interactions in addition to the LMM term. The HSI branch

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calculate the $H(\boldsymbol{A}_{\boldsymbol{X}}) \in \mathbb{R}^{n(n+1)/2\times C}$ by (6). To calculate the matrix of $\boldsymbol{B}_{\boldsymbol{X}}$, a dense layer with a nonlinear activation function is used to represent the fitting between abundance and bilinear parameters to simplify the calculation [39]. It can be formulated as

$$\boldsymbol{B}_{\boldsymbol{X}} = f_{\psi}\left(\boldsymbol{S}_{\boldsymbol{X}}\right) \tag{19}$$

where $f_{\psi}(\cdot)$ represents the nonlinear function mapping. By combining the linear and nonlinear components of the HSI branch, we can obtain the reconstructed HSI $\boldsymbol{X}_{\boldsymbol{rec}}$ by

$$\boldsymbol{X}_{\boldsymbol{rec}} = \boldsymbol{S}_{\boldsymbol{X}}\boldsymbol{A}_{\boldsymbol{X}} + \boldsymbol{B}_{\boldsymbol{X}}H\left(\boldsymbol{A}_{\boldsymbol{X}}\right). \tag{20}$$

$\boldsymbol{S}_{\hat{\boldsymbol{X}}}$ is also reconstructed through the HSI decoder to obtain LrHSI $\widehat{\boldsymbol{X}_{\boldsymbol{rec}}}$ with geometric feature alignment

$$\widehat{\boldsymbol{X}_{\boldsymbol{rec}}} = S_{\hat{X}}A_X + f_{\psi}\left(S_{\hat{X}}\right)H\left(A_X\right). \tag{21}$$

Similarly, the MSI branch uses nonlinear mapping function $f_{\sigma}(\cdot)$ to get the $\boldsymbol{B}_{\boldsymbol{Y}} = f_{\sigma}(\boldsymbol{S}_{\boldsymbol{Y}})$. Because $(\boldsymbol{AQ})\odot(\boldsymbol{AQ}) \neq H(\boldsymbol{A})\boldsymbol{Q}$, we set $H(\boldsymbol{A})\boldsymbol{Q} = \widehat{H(\boldsymbol{A})}$ and use nonlinear function mapping $f_{\delta}(\cdot)$ to get the $\widehat{H(\boldsymbol{A})}$

$$\widehat{H\left(\boldsymbol{A}\right)} = f_{\delta}\left(H\left(\boldsymbol{A}_{\boldsymbol{X}}\right)\right). \tag{22}$$

Then, according to (11), the recovered HrMSI $\boldsymbol{Y}_{\boldsymbol{rec}}$ can be restored by

$$\boldsymbol{Y}_{\boldsymbol{rec}} = \boldsymbol{S}_{\boldsymbol{Y}}\boldsymbol{A}_{\boldsymbol{Y}} + \boldsymbol{B}_{\boldsymbol{Y}}\widehat{H\left(\boldsymbol{A}\right)}. \tag{23}$$

According to (12), high-resolution abundance information from HrMSI and the endmembers extracted from LrHSI can be used to reconstruct HrHSI $\boldsymbol{Z}$. UDBFN can unmix spectral images into abundance and endmember components, adapt to more complex environments, and improve the robustness of the reconstruction.

E. Loss Function

The loss function mainly consists of four components: the reconstruction loss L_{rec}, the constraint loss L_{cons}, the sparsity loss L_{spa}, and the registration loss L_{reg}. To effectively optimize network parameters, we combine these loss functions as follows:

$$L = \alpha L_{rec} + \beta\left(L_{cons} + L_{spa}\right) + \gamma L_{reg} \tag{24}$$

where α, β, and γ serve as the tradeoff factors.

1) L_{rec} calculates the loss between the image restored by the decoder and the original input image, and $\|\cdot\|_1$ represents the ℓ_1-norm function. L_{rec} can be defined as

$$L_{rec} = \left\|\boldsymbol{X}_{\boldsymbol{rec}} - \boldsymbol{X}\right\|_1 + \left\|\boldsymbol{Y}_{\boldsymbol{rec}} - \boldsymbol{Y}\right\|_1. \tag{25}$$

2) L_{cons} indicates that the prior restrictions of abundance sum-to-one constraint (ASC). To conform to the physical meaning, at a given pixel z_j, s_i, and $b_{i,k,j}$ represent the proportion occupied by linear and bilinear components of different endmembers. L_{cons} can be written as

$$L_{cons} = \left\|1_{hw} - \left[\boldsymbol{S}_{\boldsymbol{X}}, f_{\psi}\left(\boldsymbol{S}_{\boldsymbol{X}}\right)\right]\right\|_1 + \left\|1_{HW} - \left[\boldsymbol{S}_{\boldsymbol{Y}}, f_{\sigma}\left(\boldsymbol{S}_{\boldsymbol{Y}}\right)\right]\right\|_1. \tag{26}$$

3) L_{spa} is to ensure the sparsity of the abundance matrix

$$L_{spa} = \sum_{u} D_{\mathrm{KL}}\left(\epsilon\middle\|\boldsymbol{S}_{\boldsymbol{X}}(u)\right) + \sum_{v} D_{\mathrm{KL}}\left(\epsilon\middle\|\boldsymbol{S}_{\boldsymbol{Y}}(v)\right) \tag{27}$$

where u and v index pixels in the HSI and MSI abundance maps, respectively. ϵ is a small positive target sparsity value, and $D_{\mathrm{KL}}(p\|q) = p\log(p/q) + (1-p)\log((1-p)/(1-q))$ denotes the standard KL divergence between Bernoulli sparsity targets [40]. Thus, the KL term is evaluated over pixel-indexed abundance responses.

4) L_{reg} measures the similarity between the spectral information and spatial information between the generated HSI $\boldsymbol{Z}$ and the calibrated LrHSI and also ensures that $\boldsymbol{Z}$ is aligned with the MSI features. At the same time, the consistency between the original input HrMS and LrHSI was also taken into account

$$L_{reg} = \left\|SRF\left(\boldsymbol{X}\right) - PSF\left(\boldsymbol{Y}\right)\right\|_1 + \left\|SRF\left(\boldsymbol{Z}\right) - \boldsymbol{Y}\right\|_1 + \left\|PSF\left(\boldsymbol{Z}\right) - \widehat{\boldsymbol{X}_{\boldsymbol{rec}}}\right\|_1. \tag{28}$$

where $SRF(\cdot)$ is the SRF (2) and $PSF(\cdot)$ is the PSF (3) which is composed of learnable downsampling convolutional layers.

IV. Experiments

This section reports the experimental setting and evaluation results. We first introduce the simulated datasets and deformation settings, then compare the UDBFN with representative HSI–MSI fusion baselines, and finally analyze model complexity, loss sensitivity, ablation results, and deformation tolerance.

A. Datasets

1) Simulated Datasets: We conducted experiments on three publicly available datasets: Pavia University, Washington DC, and Chikusei.

The Pavia University dataset[1] is a subset of hyperspectral data over the campus of Pavia University in Italy. Pavia University is a 610 × 340 pixel image with 115 spectral bands. After excluding 12 absorptive and noisy bands, 103 spectral bands are retained. Subsequently, a region of 256 × 256 pixels with these 103 bands is chosen as the HrHSI data for analysis.

The Washington DC dataset[2] is an aerial HSI of a shopping center in Washington, D.C., captured by the Hydice sensor. The HSI data comprises 191 spectral bands, covering the wavelength range from visible light to near-infrared (0.4–2.4 μm), with a spatial resolution of 1280 × 307 pixels. After eliminating the noisy bands, we select a 256 × 256-pixel region comprising 191 spectral bands for our experimental analysis.

The Chikusei dataset[3] was acquired on July 29, 2014. The scene consists of 2517 × 2335 pixels. We select a 256 × 256 area with 112 bands as the HrHSI data.

The simulated HrMSI is generated using the corresponding SRF data [41]. The format of the datasets is shown in Table II.

We perform rotation, translation [42], and nonlinear transformation operations on LrHSI images to simulate the

[1]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_Centre_and_University

[2]https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html

[3]https://naotoyokoya.com/Download.html

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distortion of hyperspectral and MSIs in real environments. To simulate the nonrigid deformation in LrHSI in practical situations, we generate a deformation field by linearly combining eight Gaussian distributions [18]. The nonlinear transformation can be formed as

$$\mathcal{T}(x) = x + \sum_{k} c_k \mathcal{N}_k\left(x|\mu_k, \sigma^2\right) \tag{29}$$

where $\mathcal{N}_k$ is the kth Gaussian component and c_k is the kth mixture coefficient. μ_k is the mean of the kth component and σ is the standard deviation.

We conduct comparative experiments by setting rotations of 1°, translations of 1 pixel, and nonlinear transformations of 1 pixel.

2) Quantitative Metrics: SAM [43] measures the average spectral angle between the reconstructed spectrum and the reference spectrum at each spatial location. A smaller SAM indicates better spectral fidelity. This metric evaluates spectral angular discrepancy. The specific expression is as follows:

$$SAM(I, F) = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\arccos\left(\frac{\langle I_{ij}, F_{ij}\rangle}{\|I_{ij}\|_2\|F_{ij}\|_2}\right) \tag{30}$$

where I represents the reference image, F represents the reconstructed image, and I_{ij} and F_{ij} are the spectral vectors at the (i, j)th pixel.

The PSNR assesses image quality by calculating the mean-squared error (mse) between the original and reconstructed images. A higher PSNR value indicates lower distortion, meaning the reconstructed image is closer to the original, thereby reflecting better image quality

$$MSE(I, F) = \frac{1}{CHW}\sum_{c=1}^{C}\sum_{i=1}^{H}\sum_{j=1}^{W}(I(c, i, j) - F(c, i, j))^2 \tag{31}$$

$$PSNR(I, F) = 10\log_{10}\frac{MAX^2(F)}{mse(I, F)} \tag{32}$$

where $MAX(\cdot)$ represents the maximum value of every channel.

ERGAS [44] is a global relative error metric computed from the band-wise mse normalized by the squared mean of the reference band. A smaller ERGAS indicates better reconstruction quality

$$ERGAS(I, F) = \frac{100}{r}\sqrt{\frac{1}{C}\sum_{i=1}^{C}\frac{mse(I(:,:,i), F(:,:,i))}{\mu_{I_i}^2}} \tag{33}$$

where r is the spatial scale ratio between HrMSI and LrHSI ($r = 8$ for Pavia University and Chikusei, and $r = 4$ for Washington DC), C is the number of spectral bands, and μ_{I_i} is the mean value of the ith reference HrHSI band. The mse is computed between the reconstructed HrHSI and the ground-truth HrHSI for each band.

3) Implementation Details: Our method is implemented on the NVIDIA GTX 4090 GPU based on the PyTorch framework and trained using the Adam optimizer.

In the experiment, we set the number of scale layers in HRM N as 4 on Pavia University and Chikusei. We set N as 3 on Washington DC. We set the initial learning rate to 0.005

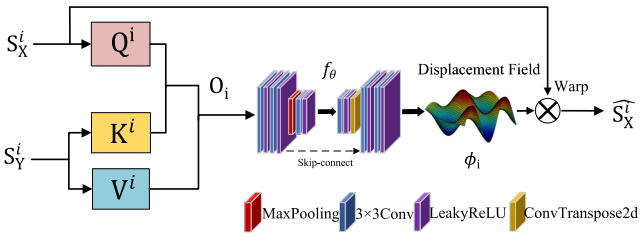


Fig. 3. Flowcharts of the CMT. The above figure shows the structure of the CMT in the ith layer.

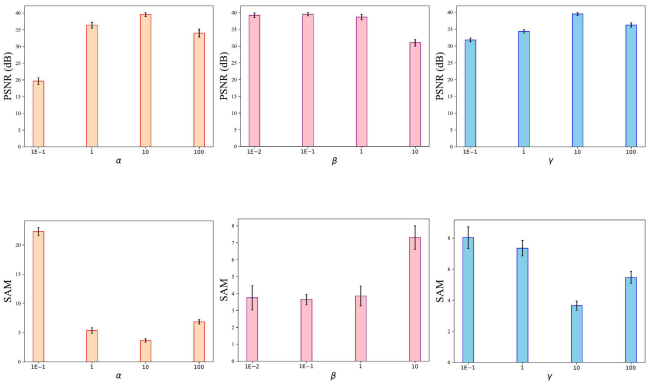


Fig. 4. Schematic of sensitivity comparison of loss function parameters α, β, and γ.

and implemented a linear step decay schedule spanning from 1000 to 4000 epochs.

About the loss function, after experimentation and comparison, we set α, β, and γ to 10, 0.1, and 10, respectively. On the well-registered Pavia University dataset, we fixed two terms, α, β, and γ, and adjusted the value of the other term for sensitivity experiments. The results are shown in Fig. 4. It can be seen that α has the highest sensitivity. When it changes from 1E-1 to 100, the values of the PSNR and SAM are greatly affected. The values of β and γ are relatively stable. The overall change in β between 0.01 and 1 is not significant. γ achieves optimal results at a value of 10, and changes in its value also have a significant impact on the PSNR, but have relatively less effect on the SAM compared to α.

B. Model Complexity Analysis

For the linear reconstruction $Z = SA$, the computational complexity is $\mathcal{O}(HW \cdot n \cdot C)$, where H and W are spatial dimensions, C is the number of spectral bands, and n is the number of endmembers. In the proposed bilinear component, element-wise interactions among endmembers yield $n(n+1)/2$ pairwise combinations, leading to a total complexity of

$$\mathcal{O}\left(HW \cdot n^2 \cdot C\right). \tag{34}$$

Although this introduces additional cost compared with the linear case, the runtime remains efficient in practice, averaging 0.415, 0.457, and 0.642 s per iteration on the Pavia University, Chikusei, and Washington DC datasets, respectively. The full training procedure of 4000 epochs requires 1273 s (≈21 min),

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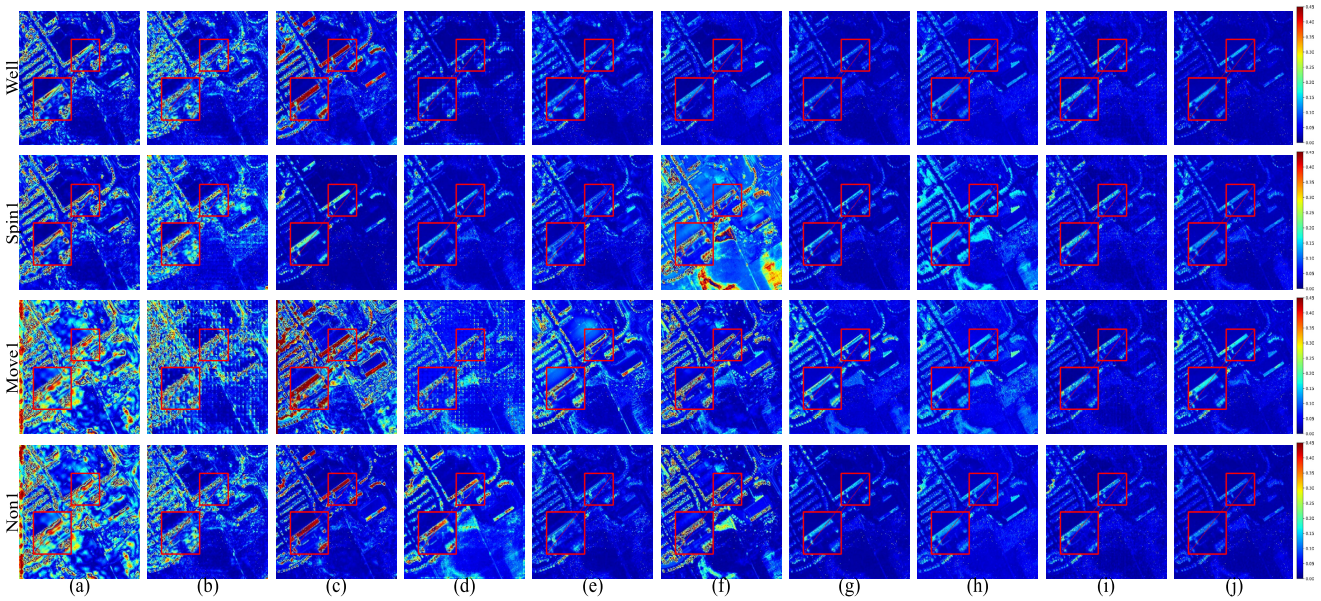


Fig. 5. Visual assessment under different deformation scenarios on the Pavia University dataset. Values in the error maps denote MRAE. "Well-regis" denotes simulated well-registered input data. "Spin1," "Move1," and "Non1" denote a 1° counterclockwise rotation, a one-pixel left shift, and a one-pixel nonrigid transformation applied to LrHSI, respectively. (a) GSA. (b) FUSE. (c) CNMF. (d) HySure. (e) PixAwa. (f) CSU. (g) CUCaNet. (h) NonRegSRNet. (i) DTDNML. (j) Ours.

TABLE III
MODEL COMPLEXITY COMPARISON

Model	Parameters	FLOPs
Ours	8.30 M	24.0G
CUCaNet	1.42 M	38.9G
NonReg	2.11 M	17.9G
PixAwa	14.05 M	204.2G
DTDNML	1.65 M	41.7G
GSA	-	12.16G
FUSE	-	13.3G
CNMF	-	9.5G
ICCV15	-	19.6G
HySure	-	23.2G

which is acceptable given the model complexity and the scale of the datasets used. This confirms that the bilinear modeling maintains manageable computational overhead relative to its performance gains.

To further evaluate the computational efficiency of the proposed UDBFN, we compare its model complexity with representative methods in terms of the number of trainable parameters, floating-point operations per second (FLOPs), and average inference time per image.

As shown in Table III, the UDBFN has 8.30 M parameters and 24 G FLOPs, achieving a practical tradeoff between accuracy and efficiency. It is much lighter than PixAwa while remaining comparable in complexity to several learning-based baselines. Compared with traditional algorithms, the UDBFN requires more computation because of the learnable registration and bilinear branches, but the runtime remains acceptable for the tested image sizes.

C. Results of Three Datasets

To comprehensively evaluate the UDBFN, we compare it with representative traditional and learning-based HSI–MSI fusion methods, including GSA [45], FUSE [46], CNMF [29], HySure [30], CSU [47], PixAwaRefin [48], CUCaNet [49], NonRegSRNet [19], DTDNML [24], and UMSFT [25], where applicable. These comparison approaches are consistent with their corresponding papers and source code. The results for the three simulated datasets are qualitatively and visually presented below. The indicator value of our experiment is obtained by taking the average of 5 experiments.

We compare the three indicators, including the SAM, ERGAS, and PSNR of various methods under complete alignment, rotation 1°, translation 1 pixel, and nonlinear deformation. These metrics jointly evaluate spectral information recovery and spatial information recovery.

1) Pavia University: First, we evaluate the visual quality of the fused images derived from the Pavia University data. The corresponding outcomes are depicted in Figs. 5 and 6 (reconstructed image) and Table IV. Table IV summarizes the quality measure results obtained using the Pavia University data under different conditions. From the first line of the image restoration results, it can be seen that under good registration conditions, CUCaNet, NonRegSRNet, and the proposed method UDBFN have better reconstruction accuracy than other methods. From the indicators and recovery results, it can be seen that CNMF has achieved relatively good results in traditional methods. However, deep-learning-based methods are generally superior to traditional methods.

For unregistered image pairs, traditional methods lacking explicit registration generally degrade under rotation and translation, especially around object boundaries. CUCaNet and NonRegSRNet are more robust than traditional baselines.

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Fig. 6. Reconstructed images on the Pavia University dataset.

TABLE IV

COMPARATIVE QUALITY ASSESSMENT ON PAVIA UNIVERSITY DATA. THE BEST-PERFORMING RESULTS ARE DISPLAYED IN BOLD

Metrics / Methods	Well-registration			Rotate 1°			Translation 1 pixel			Nonrigid Deformation		
	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR
GSA	5.476	1.632	32.789	6.785	3.368	27.384	7.894	4.351	23.854	6.312	3.548	26.784
FUSE	8.211	2.203	30.216	25.540	7.592	22.762	12.178	5.785	23.015	11.471	5.512	24.038
CNMF	3.774	1.364	34.178	15.422	5.122	26.336	11.128	3.300	29.716	4.211	1.535	33.413
HySure	3.607	0.813	35.265	4.307	1.374	31.712	8.165	1.758	33.742	4.310	1.099	33.270
PixAwa	3.635	0.623	36.480	3.753	1.075	35.009	8.743	2.875	28.825	4.105	1.214	34.213
CSU	4.275	0.899	36.980	8.246	5.294	23.900	9.876	5.893	24.925	6.346	2.039	33.581
CUCa	4.086	0.529	38.528	7.266	0.871	34.891	8.948	1.153	34.346	4.221	**0.568**	38.435
NonReg	3.568	0.493	39.907	7.121	0.834	34.479	8.178	1.649	34.795	3.998	0.743	38.253
DTDNML	3.961	1.329	38.563	4.994	1.688	36.252	5.926	1.483	34.673	4.194	1.413	37.962
UMSFT	3.905	1.173	38.694	6.406	3.460	35.344	6.164	1.945	34.507	3.782	2.167	37.926
Ours	**3.479**	**0.479**	**40.254**	**3.494**	**0.701**	**38.459**	**4.556**	**0.712**	**37.393**	**3.587**	0.614	**38.971**

TABLE V

COMPARATIVE QUALITY ASSESSMENT ON WASHINGTON DC DATA. THE BEST-PERFORMING RESULTS ARE DISPLAYED IN BOLD

Metrics / Methods	Well-registration			Rotate 1°			Translation 1pix			Nonrig Deformation		
	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR
GSA	5.375	3.217	32.305	7.315	4.502	28.856	8.165	5.168	25.676	5.568	4.287	30.432
FUSE	6.251	4.055	30.137	7.653	4.746	27.764	8.308	5.002	25.742	6.729	4.358	29.967
CNMF	2.022	1.881	38.165	2.903	3.124	37.695	8.002	4.145	29.629	2.182	2.345	37.339
HySure	6.166	4.088	31.115	8.375	4.817	27.698	10.758	5.834	23.019	6.238	3.957	31.035
PixAwa	2.947	2.945	35.416	7.011	3.980	31.743	7.218	4.165	30.651	3.849	3.385	33.687
CSU	2.248	2.347	37.410	4.307	3.827	32.564	9.197	6.387	21.982	3.913	3.918	31.588
CUCa	1.965	1.947	38.128	4.559	3.055	34.753	5.660	3.124	34.102	2.184	2.247	37.946
NonReg	**1.852**	1.803	38.899	3.047	2.993	37.579	3.732	2.235	36.208	2.064	**2.166**	38.228
DTDNML	2.309	2.659	37.291	3.128	7.191	31.929	4.147	7.734	29.908	3.062	3.182	35.129
OURS	1.965	**1.617**	**39.125**	**2.697**	**2.912**	**37.779**	**3.402**	**2.016**	**37.095**	**1.987**	2.195	**38.425**

The UDBFN obtains its clearest advantage under misregistration, where cross-modal alignment directly improves boundary preservation and spectral–spatial reconstruction. The UDBFN still leaves room for improvement in several metrics: for

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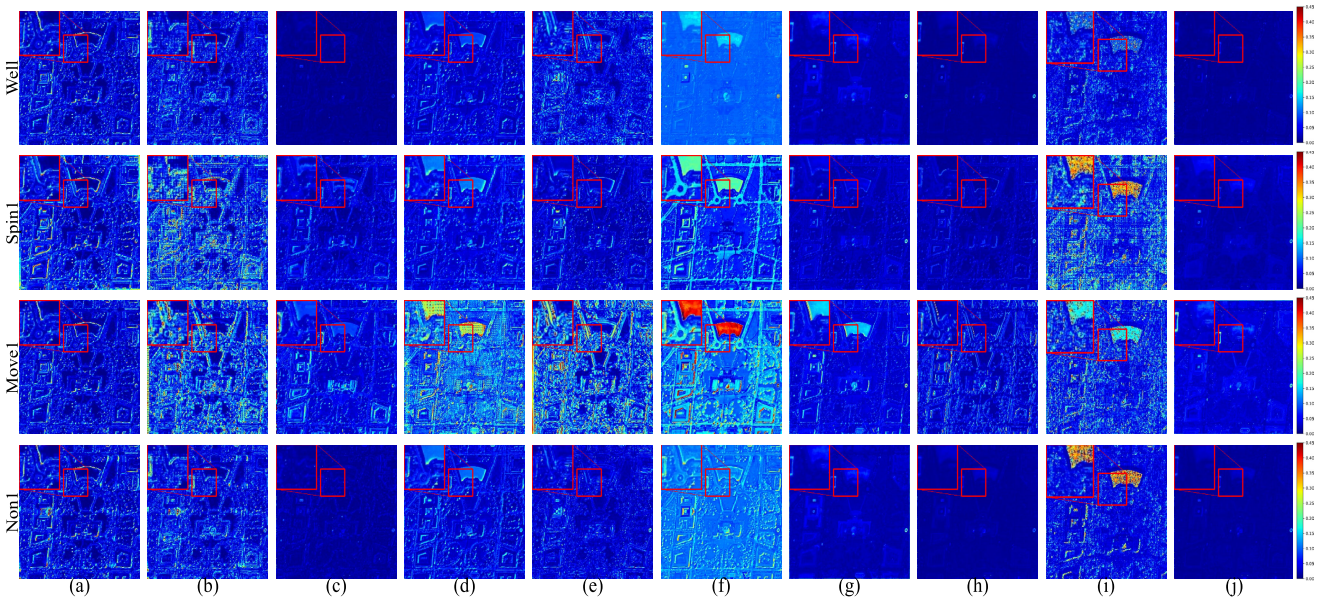


Fig. 7. Visual assessment under different deformation scenarios on the Washington DC dataset. Values in the error maps denote MRAE. Enlarged local regions are displayed in the upper-left corner for detailed comparison. (a) GSA. (b) FUSE. (c) CNMF. (d) HySure. (e) PixAwa. (f) CSU. (g) CUCaNet. (h) NonRegSRNet. (i) DTDNML. (j) Ours.



Fig. 8. Reconstructed images on the Washington DC dataset.

example, CUCaNet obtains the lowest ERGAS under the nonrigid Pavia setting. Therefore, the results should be interpreted as evidence of robustness under deformation, while unconditional superiority over all methods and metrics is left unclaimed.

2) Washington DC: The visual quality comparison of Washington DC is shown in Figs. 7 and 8 (reconstructed image), and the various evaluation indicators are shown in Table V. We evaluate the visual quality of the fused images obtained using the Washington DC data.

On the Washington DC dataset, several methods achieve competitive results under well-registered conditions because the 4× scale factor preserves more LrHSI spatial information. Under rotation and translation, however, methods without robust alignment show larger boundary errors. The UDBFN obtains the best PSNR in all four settings and the best SAM/ERGAS in most misregistered settings, while NonRegSRNet remains slightly better in SAM under well-registered conditions and ERGAS under nonrigid deformation. These results again indicate that the main benefit of UDBFN is robustness to cross-modal spatial inconsistency.

3) Chikusei: We further evaluate the methods on the Chikusei dataset. The experimental results are shown in

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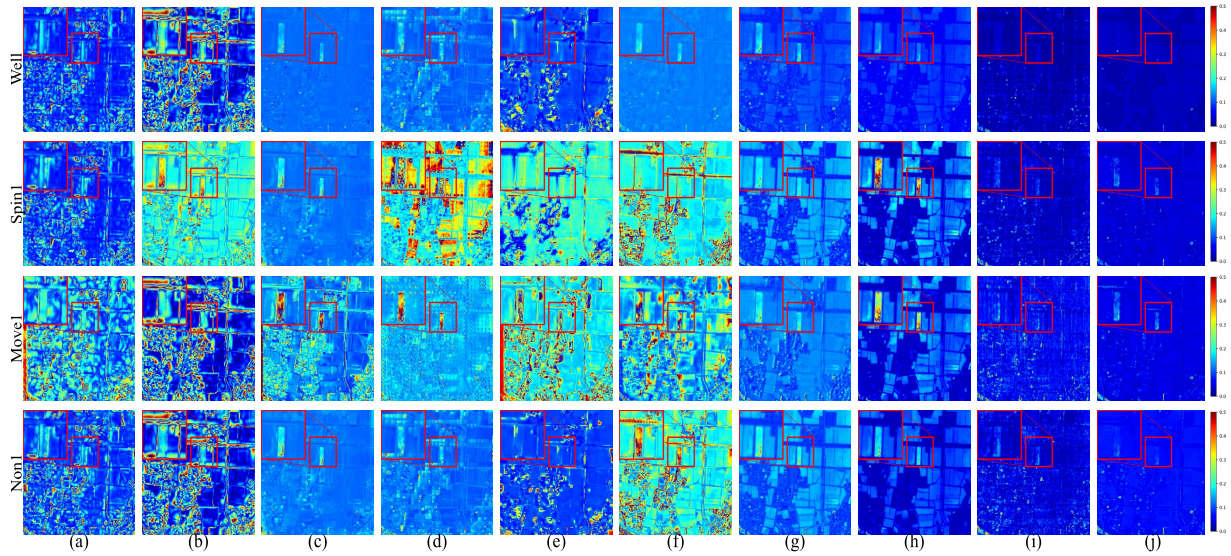


Fig. 9. Visual assessment under different deformation scenarios on the Chikusei dataset. Values in the error maps denote MRAE. (a) GSA. (b) FUSE. (c) CNMF. (d) HySure. (e) PixAwa. (f) CSU. (g) CUCaNet. (h) NonRegSRNet. (i) DTDNML. (j) Ours.

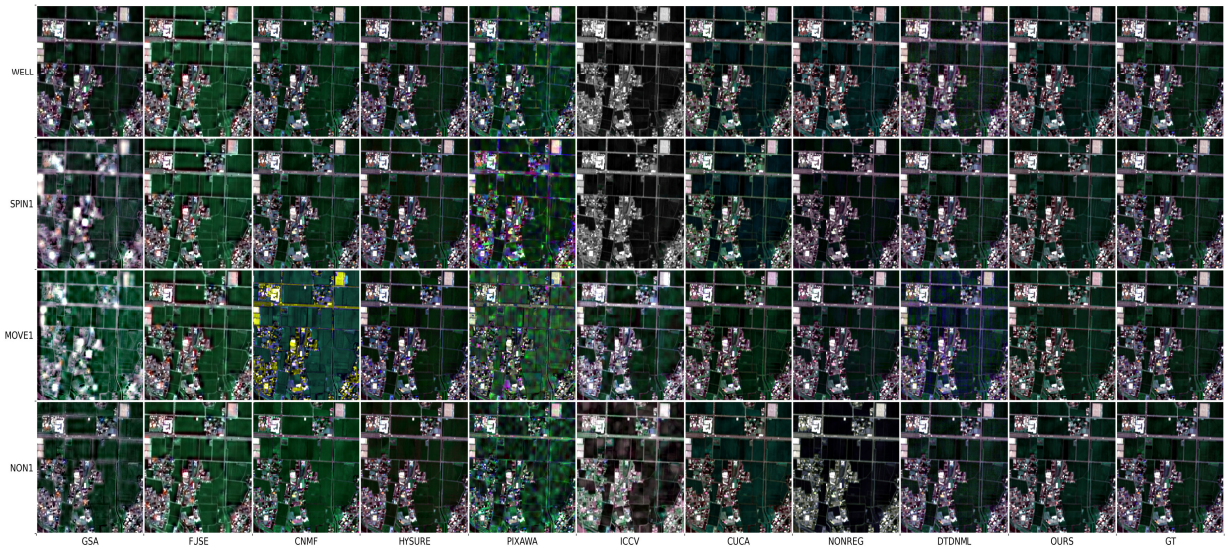


Fig. 10. Reconstructed images on the Chikusei dataset.

Figs. 9 and 10 and Table VI. Traditional methods degrade under misalignment because they rely heavily on accurate pixel-level registration. Learning-based methods are generally more stable, and the UDBFN shows clear advantages under rotation and translation. Under well-registered and nonrigid deformation, DTDNML obtains the best SAM or PSNR in some columns; accordingly, the comparison emphasizes deformation robustness and avoids a claim of uniform dominance.

D. Spectral Analysis

To further examine the bilinear formulation, we conduct a channel-wise spectral analysis on the Chikusei dataset by computing the RMSE between the reconstructed HSI and the ground truth across all spectral bands. As shown in Fig. 11, the UDBFN achieves lower RMSE across most spectral channels than the compared linear or registration-only baselines, indicating that bilinear modeling helps preserve spectral fidelity.

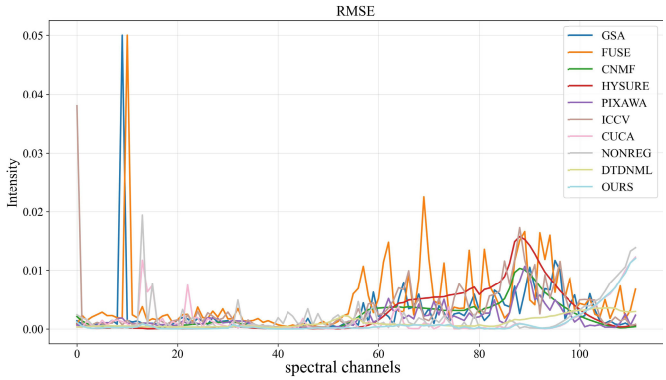


Fig. 11. Spectral analysis of the Chikusei dataset.

A slight performance drop is observed at longer wavelengths

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TABLE VI
COMPARATIVE QUALITY ASSESSMENT ON CHIKUSEI DATA. THE BEST-PERFORMING RESULTS ARE DISPLAYED IN BOLD

Methods \ Metrics	Well-registration			Rotate 1°			Translation 1pix			Nonrig Deformation		
	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR
GSA	7.108	5.261	30.747	9.584	7.322	25.700	10.538	8.134	23.245	7.553	5.788	29.227
FUSE	8.154	5.967	29.807	10.533	8.087	23.792	11.455	8.416	22.852	8.556	6.439	28.797
CNMF	2.846	3.188	36.709	6.439	5.803	29.150	6.862	6.436	28.422	2.983	3.226	36.465
HySure	4.321	4.937	31.926	6.393	7.171	26.409	6.431	7.566	25.246	4.496	5.436	30.431
PixAwa	2.453	3.517	35.478	3.639	5.354	30.312	4.117	6.440	28.369	3.498	3.624	34.974
CSU	4.457	3.468	35.787	6.553	7.642	25.166	7.611	7.447	25.684	4.523	6.472	28.426
CUCa	2.687	2.758	37.352	4.687	3.947	34.653	4.589	3.896	34.730	3.320	3.098	36.849
NonReg	2.252	2.495	38.756	3.224	3.268	36.051	2.915	3.291	36.025	2.715	2.987	37.033
DTDNML	**2.176**	2.272	**39.156**	3.221	**2.734**	36.483	3.256	3.753	33.851	**2.582**	2.654	**38.107**
OURS	2.236	**2.211**	38.971	**2.697**	3.173	**36.795**	**2.602**	**3.147**	**36.832**	2.687	**2.574**	38.060

TABLE VII
ABLATION STUDY ON LOSS FUNCTIONS

Loss Item				Well-registration			Rotate 1°			Translation 1pix			Nonrig Deformation		
L_{rec}	L_{cons}	L_{spa}	L_{reg}	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR
✓	✓	✓	✓	**3.649**	**1.297**	**39.547**	**3.566**	**1.633**	**38.023**	**4.612**	**2.179**	**37.149**	**3.745**	1.506	**38.591**
×	✓	✓	✓	8.005	3.931	27.142	7.669	4.693	26.820	10.004	5.778	22.421	8.347	4.279	26.642
✓	×	✓	✓	3.776	1.344	39.198	3.946	1.679	37.953	6.405	2.835	34.297	3.832	1.487	38.364
✓	✓	×	✓	3.654	1.309	39.407	3.864	1.702	37.821	5.678	2.677	35.814	3.825	**1.479**	38.577
✓	✓	✓	×	21.021	6.761	20.610	22.151	6.496	21.416	22.908	6.846	20.709	23.062	6.823	20.805

TABLE VIII
RESULTS OF THE MODULE ABLATION EXPERIMENT ON THE UDBFN

Module			Well-registration			Rotate 1°			Translation 1pix			Nonrig Deformation		
HRM	$BilinearC$	CMT	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR	SAM	ERGAS	PSNR
✓	✓	✓	**3.649**	**1.297**	**39.547**	**3.566**	**1.633**	**38.023**	**4.612**	**2.179**	**37.149**	**3.745**	**1.506**	**38.591**
×	✓	×	6.845	2.924	34.537	8.974	4.538	26.317	11.024	4.672	26.286	7.148	3.239	33.745
✓	✓	×	3.745	1.599	39.073	3.945	2.143	37.901	5.858	2.875	34.979	3.784	1.517	38.047
✓	×	✓	3.796	1.478	39.135	4.705	2.398	36.839	5.732	2.720	35.104	4.185	2.293	37.871

(above 868 nm), possibly because of lower signal-to-noise ratios and more complex scattering.

E. Ablation Test

1) Loss Item: To verify the effect of each loss term, we train UDBFN variants on the same Pavia University setting while removing one loss term at a time. Table VII summarizes the results.

Removing L_{rec} or L_{reg} causes the most severe degradation, showing that reconstruction fidelity and cross-modal registration consistency are essential. Removing L_{cons} or L_{spa} leads to a milder but still visible decline, especially under rotation and translation, because these terms regularize the physical range and sparsity of abundance components.

2) Encoder and Decoder Networks: To verify the effectiveness of the bilinear decoder and the alignment module, we further remove the HRM, CMT, and the bilinear component separately. Table VIII reports the module ablation results under well-registration, rotation, translation, and nonrigid deformation.

When HRM is removed, the model loses explicit cross-modal alignment and performance drops substantially. Removing only CMT also degrades the results, confirming that cross-modal interaction is useful for displacement estimation. Removing the bilinear component reduces spectral reconstruction quality, which supports the value of modeling nonlinear mixing in addition to the linear component.

3) Tolerance Study: In addition to the originally reported deformation settings (Spin1, Move1, Non1), we have conducted a series of extended experiments to further analyze the relationship between deformation intensity and model performance.

Specifically, we introduce the following settings.

1) *Rotational Deformation:*2°–6° counterclockwise rotations (Spin2—Spin6).
2) *Translational Shifts:* 2-pixel and 3-pixel displacements (Move2–Move3).
3) *Nonrigid Deformations:* Using 3 and 5 Gaussian-based displacement fields (Non3–Non5).

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TABLE IX
PERFORMANCE OF THE UDBFN UNDER DIFFERENT DEFORMATION INTENSITIES

Deformation	PSNR ↑	SAM ↓	ERGAS ↓
Spin2 (2° rotation)	36.198	5.170	1.497
Spin3 (3° rotation)	35.041	6.403	1.761
Spin4 (4° rotation)	35.311	6.189	1.706
Spin5 (5° rotation)	35.852	5.299	1.679
Spin6 (6° rotation)	34.330	6.218	1.968
Move2 (2 px shift)	32.164	9.102	2.763
Move3 (3 px shift)	28.291	11.413	3.478
Non3 (3–kernel warp)	39.023	3.729	1.144
Non5 (5–kernel warp)	36.999	4.568	1.349

Table IX summarizes the PSNR, SAM, and ERGAS values under each deformation level. Performance generally degrades as deformation intensity increases. Translational shifts have a more severe impact than small rotations, while the model remains relatively robust under moderate nonrigid distortions. These results also indicate a limitation: very large misregistration remains challenging and will require stronger alignment constraints or coarse-to-fine matching strategies.

V. CONCLUSION

This article presents the UDBFN, an unsupervised fusion-based HSI-SR method for misregistered LrHSI and HrMSI. The proposed framework combines hierarchical abundance-domain registration, cross-modal displacement estimation, and bilinear spectral decoding. Experiments on three simulated benchmark datasets show that the UDBFN is especially effective under rotation, translation, and nonrigid deformation, while remaining competitive under well-registered settings. The method is still limited when the misregistration becomes large, and the bilinear branch increases computational cost for large images. Future work will focus on more efficient coarse-to-fine registration and lighter nonlinear unmixing modules.

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